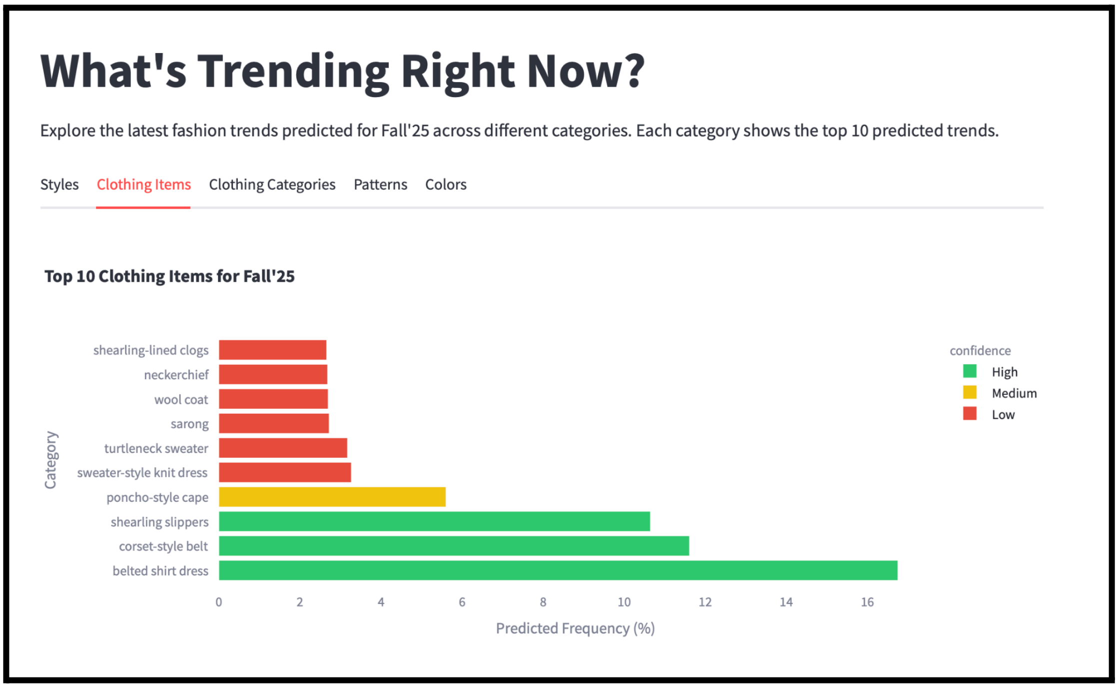

Leveraging CNNs and Time Series for Optimizing Retail Fashion

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[Github Link]

1. Introduction

The fashion industry has undergone a dramatic transformation driven by heightened consumer interest, the influence of social media, and rapid technological advancements. This evolution is largely fueled by the constant circulation of new trends and viral images, which can place a trending item in a consumer's shopping cart within seconds. However, despite this growth, the fashion industry continues to face volatility. This is due to unpredictable consumer behavior and the challenges brands face in keeping up with rapidly shifting trends. Artificial intelligence offers a promising solution to these challenges. According to McKinsey's 2025 Fashion Report, "AI-powered curation, content, and search can help customers discover brands and products more effectively—and feel more inclined to make a purchase." Traditionally, trend identification has relied heavily on human intuition, historical sales data, and industry reports. While valuable, these manual methods are slow, prone to technical errors, and struggle to match the speed and scale of modern fashion cycles. Especially in an atmosphere where an item can fall from loved to fad in as little as four months.

This paper proposes a technical approach to trend forecasting that leverages recent advancements in computer vision and machine learning. We demonstrate how Convolutional Neural Networks (CNNs) can be used for visual classification tasks such as apparel recognition, texture detection, and image segmentation. To complement this, we incorporate an LSTM to model the evolution of trends over time. This would allow brands to identify behavioral patterns and correlate them with market response and consumer engagement.

By bridging CNN-based image analysis with time series forecasting, this project aims to empower fashion retailers to respond to consumer demand in real-time by bridging CNN-based image analysis with time series forecasting. Our goal is to close the gap between the runway and the retail shelf, enabling brands to deliver timely, data-driven products to consumers.

2. Literature Review

Fortunately, significant groundwork has already been laid in the development of automated systems for fashion recognition. In building our own models, we drew inspiration from two technical projects: DeepFashion Dataset and Where To Buy It. Both allow leverage deep learning and computer vision techniques, providing valuable insights into how fashion items can be effectively identified, classified, and retrieved. These contributions offered a comprehensive view of what is technically possible and informed the direction of our approach.

One of the most influential benchmarks in fashion computer vision is DeepFashion, a large-scale dataset comprising over 800,000 images annotated with more than 1,000 attributes, 50 clothing categories, and detailed landmarks for precise localization. The creators of DeepFashion aimed to reflect the complexities of real-world fashion data, emphasizing that "Clothes often have large variations in style, texture, and cutting... frequently subject to deformation and occlusion... [and] exhibit serious variations when taken under different scenarios, such as selfies vs. online shopping photos."

To address these challenges, the DeepFashion project introduced FashionNet, a novel CNN architecture designed to jointly predict clothing attributes and landmarks. The model was specifically tailored to handle occlusions and deformations through a specialized pooling mechanism based on predicted keypoints. This learning approach has influenced many subsequent efforts in fashion analytics by highlighting the need for systems that are robust to non-standard poses, varied lighting, and background noise.

In the context of this project, DeepFashion serves as a foundational reference for building CNN-based pipelines that can extract reliable style features from complex visual data, such as runway images. While our dataset focuses on curated, high-quality imagery from fashion week runways, similar challenges like identifying silhouettes and attributes across varied poses and lighting still remain and justify the use of advanced deep learning models.

Complementing DeepFashion's dataset-centric approach, the Where To Buy It project tackled the complex "Exact Street to Shop" retrieval problem. Unlike traditional image retrieval methods that focus on stylistic similarity, WTBI sought to match real-world images of clothing with exact counterparts in online store inventories. The authors introduced a deep similarity network trained on over 39,000 exact match pairs, significantly improving retrieval accuracy by learning domain-bridging metrics between the noisy, unconstrained street images and the clean, high-resolution shop photos. Their method highlighted the challenge of domain adaptation in image retrieval and the importance of learning fine-grained, category-specific similarity functions

Another important work is that of FashionCLIP (Chia et al., 2022). This open source model demonstrated zero-shot learning for fashion concepts using CLIP models, enabling recognition without extensive labeled data. We have relied on this past work to generate our labelled dataset for LSTM predictions.

Together, these works provided a foundational framework for integrating convolutional neural networks with temporal or sequence-based modeling, particularly for time-series fashion trend analysis. While DeepFashion demonstrated the effectiveness of spatial modeling through landmark-guided CNNs, WTBI emphasized the role of domain-aware similarity learning. Our approach seeks to synthesize these insights, employing CNNs not only for static image recognition but also in combination with temporal data modeling to identify evolving fashion trends and enhance cross-temporal retrieval performance.

3. ML Methodology and Data Collection

As aforementioned the project implements a dual-module machine learning system combining deep learning for visual feature extraction and time series analysis for trend forecasting. The objective is to detect emerging fashion trends from runway imagery and correlate them with public interest using search data. The end goal is to inform strategic decisions for online retailers by connecting visual fashion cues to temporal consumer behavior.

3.1 Data

Our primary dataset consists of high-resolution fashion runway images scraped from the NowFashion Fall 2025 collection. These images capture the latest runway presentations, offering an updated and rich source of fashion-forward styles across designers and global trends. This can be located here: Fall 2025 Runway Collection.

To extract these images we employed web scraping tools built in Python (using requests, BeautifulSoup, and Selenium) to download and organize image assets by designer and collection. While this dataset provides an up-to-date snapshot of fashion innovation, its lack of labels required the use of the FashionClip model for labelling it. We used X API to provide us with the latest trending jargon regarding fashion styles, patterns, categories and more. For example, “sherling”, “faux fur”, “halter neck tops” and so on. We were able to enhance the predefined feature taxonomies for the FashionClip model and create a comprehensive labelled dataset for seasons spanning from Fall’23 to Fall’25.

For the training phase, we utilized the "Fashion Product Images" dataset from Kaggle. This dataset includes thousands of labeled clothing product images. This dataset contains annotations for both clothing categories (e.g., T-shirts, hoodies, trousers) and dominant colors (e.g., red, black, green). It enabled us to build and validate a deep learning model capable of generalizing from e-commerce-style imagery to more complex, real-world runway images.

3.2 CNN Methodology

To process and classify the visual data, we developed a custom Convolutional Neural Network using PyTorch. The model employs a multi-task learning architecture with two output heads: one for predicting the clothing article type and another for identifying the dominant color. The shared base of the network includes three stacked convolutional layers that extract increasingly abstract visual features, two max-pooling layers to downsample the spatial resolution while retaining key patterns, a dropout layer to mitigate overfitting, and two fully connected dense layers that support high-level reasoning before branching into the classification heads.

Input images were resized to 64×64 pixels, normalized, and augmented with random horizontal flips and crops to improve the model's generalization ability. During inference, the model outputs both top-1 and top-3 predictions for each classification task. Performance was measured using accuracy scores and confusion matrices. To enhance model interpretability, we implemented Grad-CAM visualizations, allowing us to verify that the CNN's predictions were grounded in the clothing features themselves rather than background elements. This step was particularly important given the variability in runway photography, which includes inconsistent lighting, camera angles, and poses.

3.3 Time Series Methodology

To accompany the visual analysis, we developed a time series forecasting pipeline to track public interest in fashion trends. This system uses the gtrendsR package in R to fetch normalized weekly search interest data from Google Trends between 2019 and 2024. We focused on keywords directly related to the clothing categories detected by the CNN, including terms such as "baggy jeans," "cargo pants," and "leather jacket." The collected data was cleaned and exported to CSV files for reproducibility and long-term analysis.

To analyze the temporal patterns in search interest, we conducted classical time series decomposition. This allowed us to separate each signal into three components: the long-term trend, the seasonal component (capturing repeating annual patterns, such as spikes during fashion weeks), and the residual noise. Visualizations were created using ggplot2 to highlight fluctuations in public interest over time. This component of the project enables validation of the CNN outputs by revealing whether runway trends coincide with or predict spikes in consumer attention.

3.4 LSTM Implementation

To forecast future trend trajectories, we implemented a Long Short-Term Memory (LSTM) model—a form of recurrent neural network well-suited for handling the complexities inherent in time series data. The LSTM model was employed to:

- **Predict Future Trends:** By learning from the historical data, the model generates forecasts of future search interest trajectories.
- **Validate Against Actual Data:** The model's predictions were subsequently compared with actual data from Fall '25, thereby enabling us to assess the accuracy and reliability of the forecasts.

- **Estimate Confidence Levels:** In addition to the predictions, the model provides confidence metrics, which offer insight into the certainty associated with each forecast and help identify potential areas of uncertainty.

Together, the image classification model and the trend forecasting system provide a holistic pipeline for analyzing the lifecycle of fashion trends from runway to the public. This integrated approach achieves the following:

- **Validation of Runway Trends:** We are able to assess whether the trends showcased on the runway accurately predict consumer search behaviors and, by extension, market trends.
- **Lifecycle Tracking of Fashion Trends:** Our system tracks the evolution of fashion trends from their initial introduction through their eventual mainstream adoption, providing a comprehensive understanding of trend lifecycles.
- **Actionable Business Insights:** The insights derived from our analysis inform critical decisions related to inventory management, product design, and strategic planning, thereby fostering data-driven fashion forecasting.

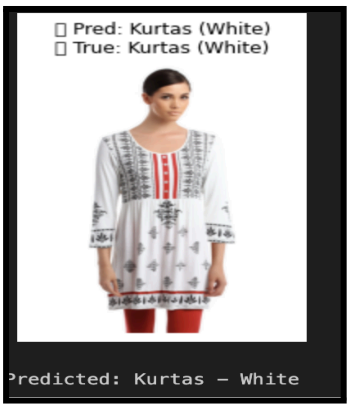
4. ML Results and Interpretation

4.1 Fashion Image Classification Using CNN

After implementing a custom CNN in PyTorch for multi-task classification of fashion product images, we evaluated its performance on the test set. The model was trained for 10 epochs on resized 64×64 RGB images and featured a shared backbone with two dense output heads for predicting Article Type and Dominant Color. Using cross-entropy loss for both tasks, the CNN achieved 65.25% accuracy on Article Type classification and 42% accuracy on Dominant Color prediction.

These results indicate that the model was moderately successful at capturing structural features (e.g., shape, sleeve length, garment outline) but less effective at recognizing colors, likely due to multiple dominant tones per image, label noise, or image lighting variation.

Figure 1 shows a representative image from the test set, in which the model accurately predicted clothing type (Kurtas) and dominant color (White).

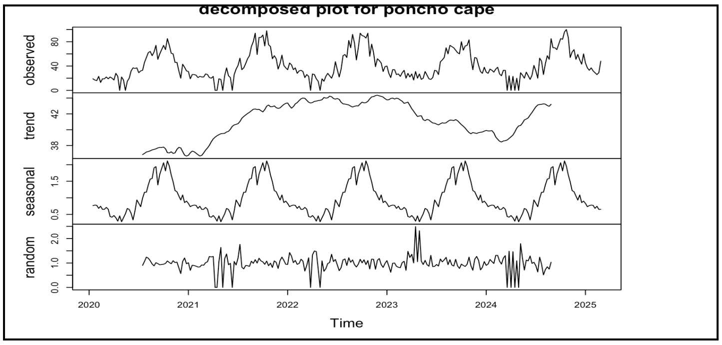


4.2 Trend Forecasting with Google Trends Data

To complement our CNN's visual classification, we performed a time series analysis using Google Trends data for four fashion-related keywords: *turtleneck sweater*, *long-sleeve bodysuit*, *shearling slippers*, and *poncho cape*. We tracked weekly search interest over a 5-year period to uncover shifts in consumer demand and recurring seasonal behaviors. Using classical multiplicative decomposition in R, we separated each time series into three key components:

- Trend – captures long-term changes in popularity over time
- Seasonal – reveals consistent, cyclical patterns that repeat annually
- Residual – accounts for random noise and short-term irregularities

This decomposition provided deeper insight into how interest in each item evolves, offering valuable context to support our image-based predictions.



In Figure 2 the term "poncho cape" exhibits a steadily rising trend from 2020 through early 2025. It also shows clear seasonal spikes, particularly in late Q4, which align with fall and winter fashion cycles. Additionally, the time series has relatively low noise, indicating that the pattern is highly predictable.

4.3 Comparing All Keywords Over Time

To identify broader seasonal fashion cycles, we visualized all four trendlines together.

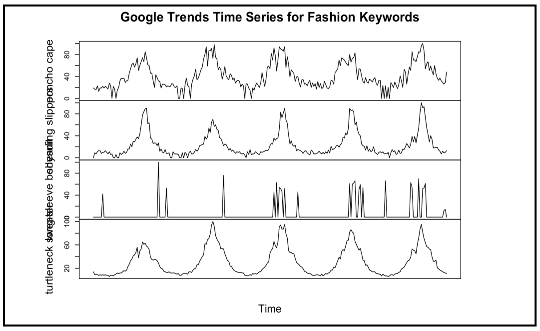


Figure 3 reveals several notable insights. Both "turtleneck sweater" and "shearling slippers" displayed clear seasonal peaks each winter, indicating strong cold-weather demand. In contrast, the "poncho cape" showed a compound pattern of seasonality combined with an upward trend over time. Meanwhile, the "long-sleeve bodysuit" exhibited less consistent patterns and higher signal noise, suggesting irregular usage or a more niche market appeal.

5. Conclusions and Future Work

5.1 Interpretation and Project Synthesis

Together, the image-based CNN and trend-based time series analysis provide a dual-perspective model for understanding fashion dynamics:

- CNN predicts garments' visual attributes, which can be used for content tagging, automated inventory labeling, or consumer search filtering.
- The time series analysis reveals macro-level demand cycles that can inform when certain styles are likely to surge in public interest.

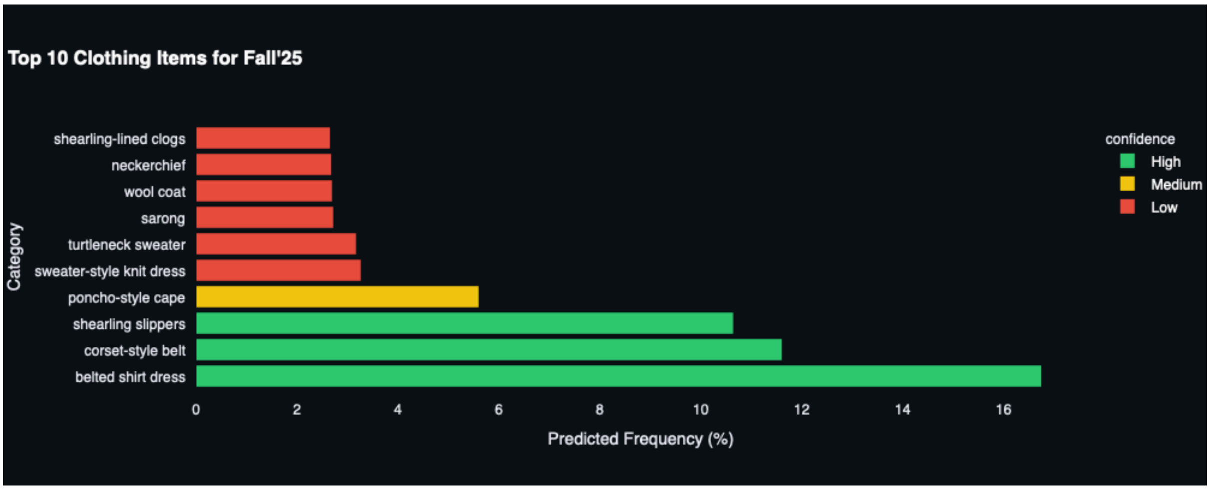


Figure 4: Prediction About Clothing Items as shown on our Streamlit Dashboard

This hybrid approach enables data-driven fashion forecasting from the supply (visual product data) and demand (consumer search behavior). This also allows for a target audience of online ecommerce retailers to utilize our tools to get the most relevant clothing that their consumers would want to see. This produces not only for them to gain more traction on their website but also allows users to stay hooked and end up buying from their online platform. The forecasting has a duel focus not only is it able to be used as a way for retailers to know what to keep on the top of their screens but also it can be use for consumers to understand what they should style their clothes with for a certain season or year.

5.2 Limitations

Throughout our project we faced a handful of limitations. Working with a neural network and hundreds and thousands of images can be taxing on our laptops. Not having access to a PC harmed our runtime and ability to create edits to the neural network and forecasting model in a timely manner. Beyond the taxation the computation had, there were also api calls to the google trends that ended up getting blocked because of too many calls and our IP being banned from accessing google trends so we had to manually gather the data into a csv to then create the time series analysis. These limitations led us to be set back time wise throughout our project and having difficulties with finding the right dataset to encompass all the different types of garments and colors is also a limiting factor throughout our project.

5.3 Future Improvements

Moving forward with our project we have multiple pathways to go about to improve our models. We can start by changing our preprocessing techniques by isolating dominant color via OpenCV may benefit color classification. Alternatively, having two separate CNNs rather than one for both classifications. We could also look into data augmentation by horizontal flipping and color jittering which could improve generalization. Due to the lack of time we could look into a longer training schedule (e.g., 100–500 epochs) would likely yield better convergence. The use of class weights could balance the effect of underrepresented clothing types. Similarly, having a more comprehensive LSTM would increase the accuracy of our predictions.

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